

## ❖ **Application Overview: E-Commerce Website**

- **Introduction:-**

- Users can use this e-commerce software like any other e-commerce site, such as to view items, add items to a shopping cart, check out, and order. It also has an admin dashboard where to manage the products, see the orders and add or remove users. The application is developed with PHP and MySQL for the back-end and HTML, CSS, and JS for the front-end .
- A description of the use, setup, and important features of the application is presented.

- **Functionality :-**

- The application is meant for customers to shop online and for administrators to deal with the back-end easily. Below, the core user and administrator capabilities are described:

### **1. User (Customer) Functionality**

- **Product Browsing:** A list of products is displayed on the application's homepage. The information for each product includes the name, price, description, and an image. Consumers can then browse the products and select to purchase them.
- **Shopping Cart:** After a user chooses products, he or she can add them to a shopping cart. The shopping cart contains the chosen products that display the items' names, prices, quantities, and an overall sum for the products. Products can be edited or deleted by the user.
- **Checkout-** When ready to purchase, the user is taken to a checkout page. Users enter their personal information on this page (name, email, shipping address, phone number). They can see their order, the final price, and approve it before paying. After the order is placed the "order confirmation page" is generated which summarizes the users order.

- **Order Confirmation** : Once the order has been completed the user gets a order confirmation including their order id, purchased items, total due, shipping address, and customer information.

## 2. Admin Capabilities

- **User Control:** Admins can add new individuals into the system through a secure Admin Dashboard. This is done through username and password specification. Currently, the implementation is storing the passwords using a basic MD5 hash. In production environments, it is better to use `password_hash()`.
- **Product Control:** Admins can include new items into the catalog. Each item is described by an image, a product name, price, description, stock quantity, etc. The items are stored in the database and made visible to the customers on the homepage. The products may be updated or removed by the Admins.
- **Order Access:** Admins have access to all orders made by users. This includes specifics of the product being ordered, their quantities, and particular details of the customer.

- **System Initialization:**

- To set the e-commerce application up and run it, adhere to the following instructions.

- **Initial Requirements:**

- **Web Server:** The application is supported by a web server which is PHP enabled. For example, Apache or Nginx.
- **Database Server:** Product details, user accounts, and orders need a MySQL database to store them.
- **PHP:** Make certain that PHP is configured onto your server as well.
- **File Permissions:** Make certain the server has proper file permissions which allow uploading images and allowing writing to those directories.

- **Step-by-Step Setup:**

1. **Download the Application Files:**

- Get the application source code files.
- All PHP scripts, HTML, CSS, and image files should be included.

## **2. Set Up the Database:**

- Load the supplied database schema into your MySQL server. The schema is available either as a dump file or as SQL queries.
- In the shopping\_cart\_db database, create the required tables, for example products, orders, users, order\_items.

## **3. Configuration:**

- Next, open the db.php file and set your database connection's host, username, password, and database name.
- Configure file permissions for image uploads; the server directory for uploaded images (e.g. the images/ folder) must be writable .

## **4. Start the Web Server:**

- Once the database has been set up and the files in place, start the web server (Apache, Nginx, etc.).
- Make sure to point the server to the app's root directory.

## **5. Accessing the Application:**

- Open your browser and go to your server's address, for example: http://localhost.
- To access the admin panel, login as an administrator, which you can configure to do by accessing the users table of the database.

# **• Key Features**

## **1. Responsive Design:**

- The application is responsive and therefore usable on desktops, tablets, and smartphones. Basic CSS formatting is applied on the front end to guarantee users have a uniform and seamless experience on varying screen sizes.

## **2. Product Management:**

- Products can be added by the admin user along with product name, description, price, stock quantity and product images.

- Products may be uploaded, so that customers may see the products and add them to their cart. Products are editable and deletable by an admin at any time.

### **3. Order Management:**

- The admin can see all customer orders, which includes the customer's name, contact information, shipping address, and items purchased. Orders are associated with a specific ID for tracking purposes.

### **4. Authentication and Authorization:**

- It identifies admin vs. regular users based on session auth. The admin dashboard is only visible to logged in users with admin privileges. Unauthorized users are then browsed to the login page.

### **5. Image Uploads:**

Admins can upload images of products from within the application and are presented with the images alongside the product details . This allows customers to see what they are buying.

### **6. Order Confirmation and Receipt:**

After a user creates an order, they receive a confirmation message with the order details including purchased product, purchased quantity, price, and total.

## **❖ Conclusion**

It is a simple, yet functional, e-commerce application that allows customers to shop online and has a user-friendly administrator's dashboard for managing users, products and orders. As users can simply install the application, thus can start shopping right away, while administrators can have total control of the back-end. This application provides everything needed to create an online store including a secure users management system, a product catalog and a responsive design.